

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I V. Sree Ruthin (192324112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sree Ruthin

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is

typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable

to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

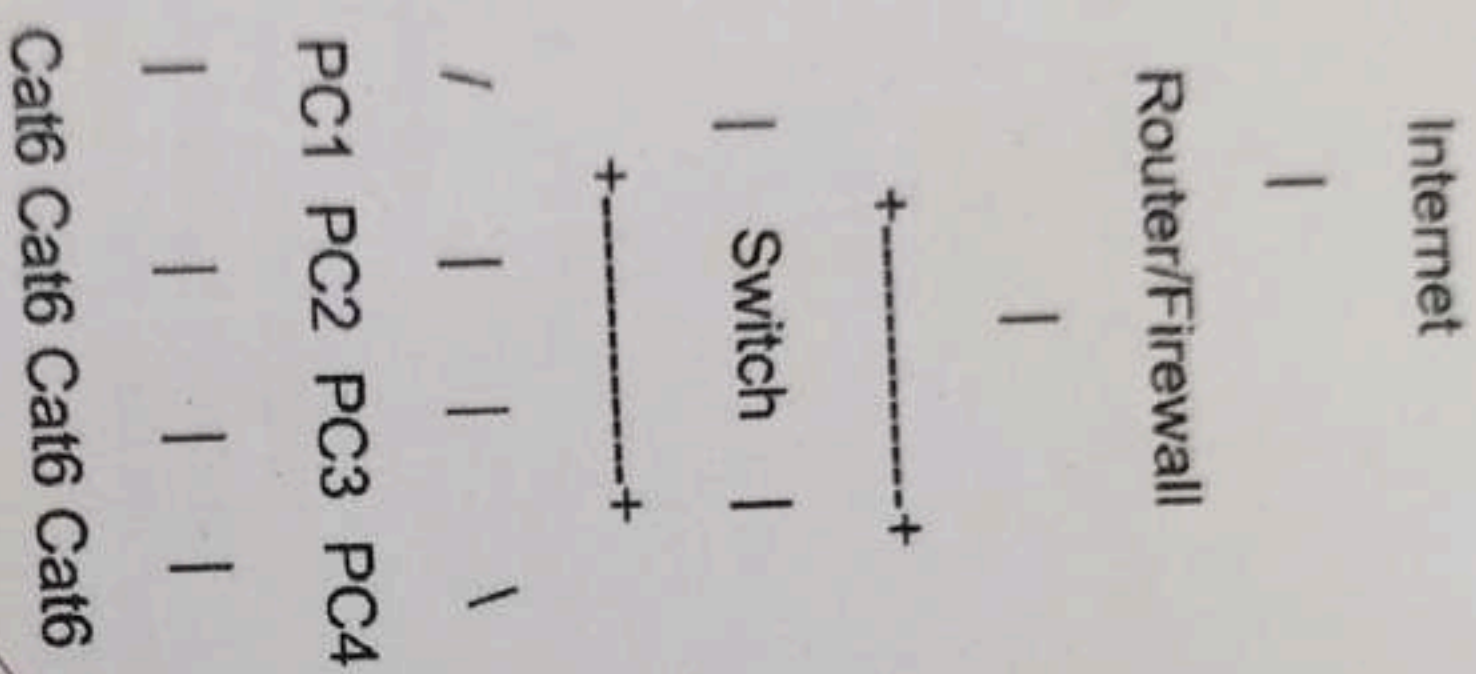
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to

1. 20-Person Startup — Star Topology

Problem:
A startup needs a simple, affordable, manageable network for around 20 employees, including computers, printers, VoIP phones and wireless devices.

Solution:

Proposed
Use a star topology with a managed 24/48-port switch at the center.



Technologies:

- Managed Layer 2/Layer 3 switch
- Cat6 UTP
- Router/firewall
- Wi-Fi access point
- DHCP
- NAT
- Gigabit Ethernet

Justification:

If one cable or workstation fails, the other devices continue operating. Centralized management makes troubleshooting and future expansion easy.

Validation:

Disconnect one workstation cable and verify that the remaining devices continue communicating.

2. Bank — Mesh Topology

Problem:

A bank requires extremely high availability because network failure can affect financial transactions and services.

Proposed

Use a **partial/full mesh topology** at the core with redundant routers, switches and links.

Solution:

Router 1

/ \

/ \

Core SW1 ---- Core SW2

| \ / |

| \ / |

| \ / |

Firewall Firewall

Technologies:

- Fiber optic cables
- Redundant routers
- Redundant core switches
- OSPF/BGP
- Firewall
- UPS
- RAID
- Single-mode/multi-mode fiber

Justification:

If one link fails, traffic can use another available path. This provides high availability and minimizes downtime.

Validation:

Disconnect one core link and verify that traffic continues through the alternate route.

3. Classroom Lab — Bus Topology

Problem:

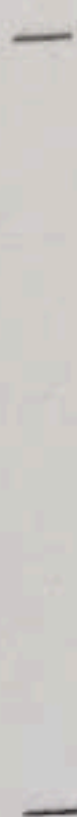
A small classroom requires a low-cost network with limited hardware requirements.

Proposed

A traditional **bus topology** can be used where appropriate, using a common backbone cable.

Solution:

Terminator Terminator



+---- PC1 ---- PC2 ---- PC3 ---- PC4

Common Bus Cable

Technologies:

- Coaxial cable
- BNC connectors
- T-connectors
- 50-ohm terminators

Advantages:

- Low hardware cost
- Simple design
- Suitable for small/light workloads

Limitations:

- Backbone failure affects the network
- Collisions can occur
- Difficult to troubleshoot as the network grows

Validation:

Test communication between multiple computers and observe the effect of a backbone cable failure.

Note: For a modern classroom network, a switched **star topology** using Cat6 and an unmanaged switch is generally more practical than a traditional bus network.

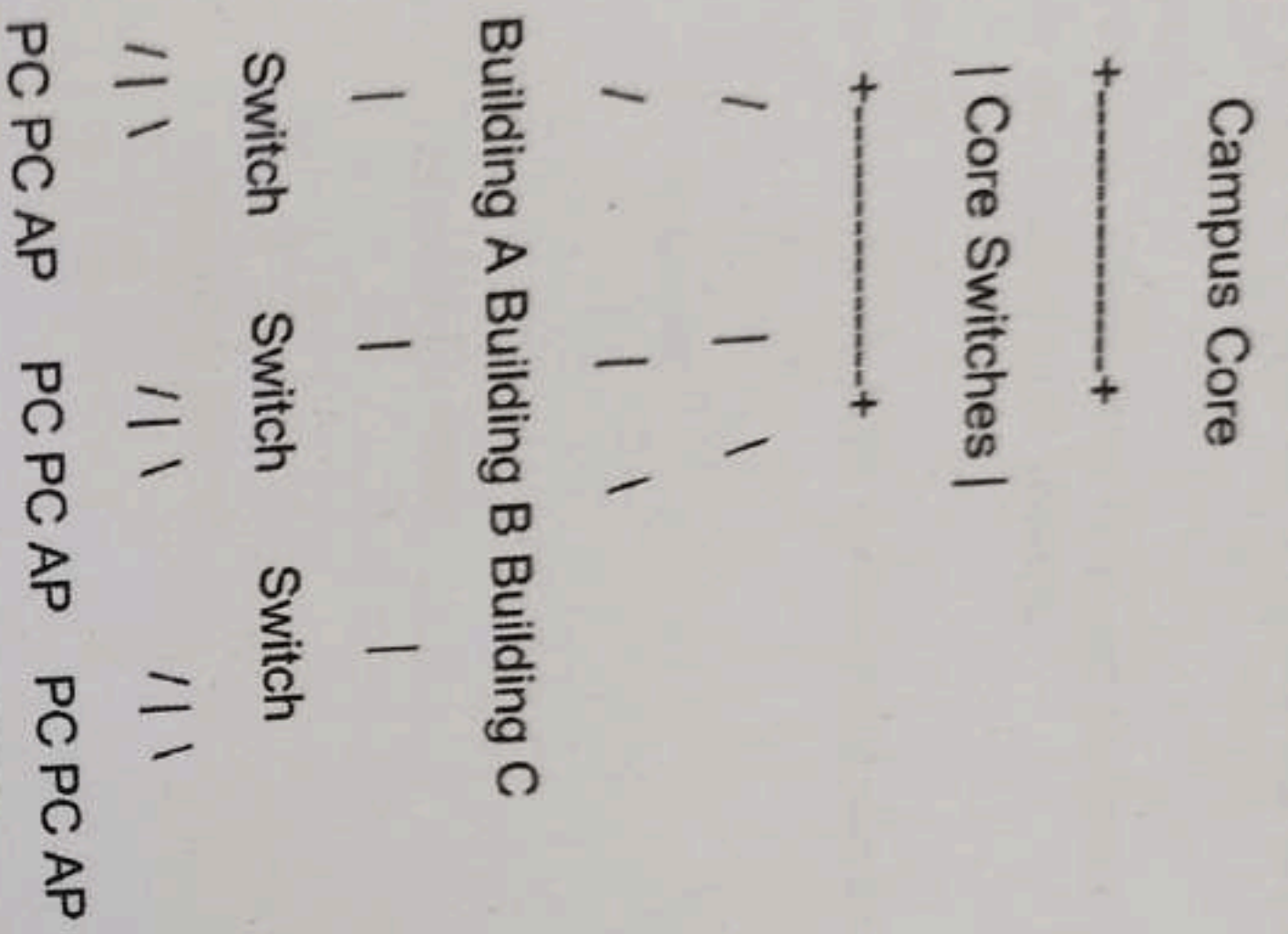
4. University Campus — Hybrid Topology

Problem:

A university has thousands of users distributed across many buildings, so one topology cannot efficiently serve the entire campus.

Proposed
 Use a **hybrid topology** combining mesh/ring at the backbone and star topology within buildings.

Solution:



Technologies:

- Fiber optic backbone
- 10/40 Gbps links
- Layer 3 switches
- Cat6
- Wi-Fi access points
- PoE
- VLANs
- Core/distribution/access architecture

Justification:

The backbone provides redundancy while star topology inside buildings makes management easy. New buildings can be added without redesigning the complete network.

Validation:

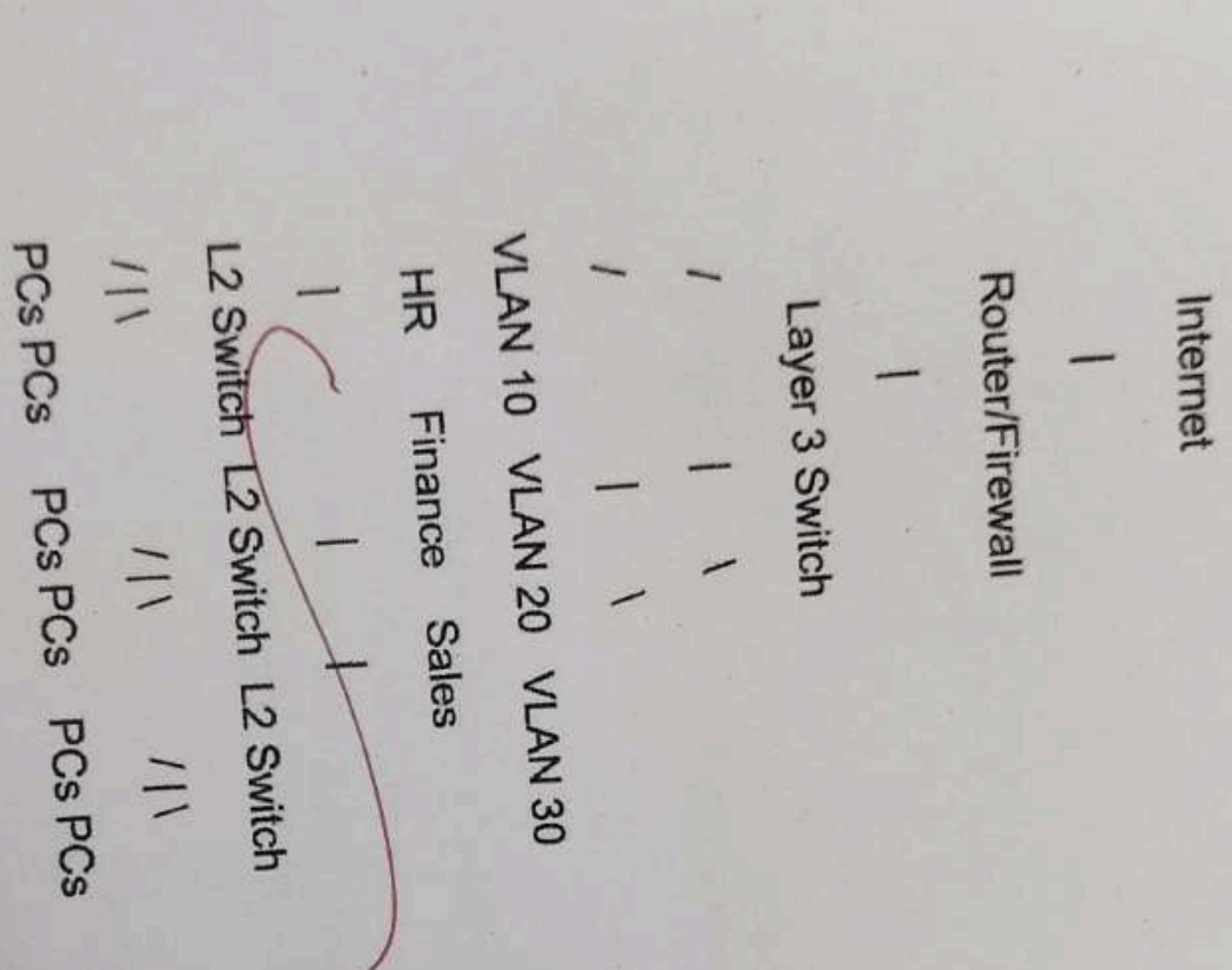
Disconnect one fiber uplink and verify that the building remains reachable through its redundant path.

5. Company with Multiple Departments — VLAN-Based Hierarchical Network

Problem:
HR, Finance, Sales and IT need to communicate while keeping departmental traffic logically separated and secure.

Proposed
Create separate VLANs for each department and use a Layer 3 switch for inter-VLAN routing.

Solution:



Example VLAN Configuration

Departmen	VLAN	Example Network
HR	10	192.168.10.0/24
Finance	20	192.168.20.0/24

Sales	30	192.168.30.0/24
IT	40	192.168.40.0/24

Technologies:

- Layer 2 switches
- Layer 3 switch
- VLANs
- Inter-VLAN routing
- Router/firewall
- DHCP
- Subnetting
- Redundant links

Justification:

VLANs isolate departments, reduce unnecessary broadcast traffic and improve security. Inter-VLAN routing allows authorized communication between departments.

Validation:

Test

1. HR → HR communication should work.
2. HR → Finance should pass through Layer 3 routing.
3. Unauthorized departmental traffic should be blocked using ACL/firewall rules.
4. Verify DHCP assigns addresses from the correct subnet.